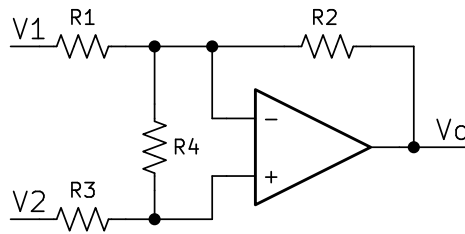
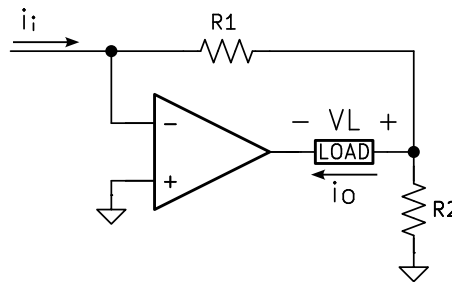


Q.1 (50 pnts): Answer the subquestions below with respect to the circuit on the right. Assume that $r_d = \infty$, $r_o = 0$, and $\alpha \neq \infty$ unless otherwise stated or implied.



- i) Find the **ideal** V_o with respect to the inputs V_1 , V_2 . Simplify. (10).
- ii) Calculate the return ratio, T . (15)
- iii) With $V_2 = 0$ and $r_o \neq 0$, find a_{ft} . (10)
- iv) With $V_1 \neq 0$, $V_2 \neq 0$, and $r_o = 0$, find the non-ideal V_o . Hint: Employ superposition using the quantity(ies) found above. (5)
- v) From i): Determine the differential mode and common mode gains, A_{DM} and A_{CM} . Simplify. (5)
- vi) From v): What is the CMRR for this circuit? Propose values for the resistors which produce a CMRR of 40 dB. (5)

Q.2 (50 pnts): Answer the subquestions below with respect to the current amplifier circuit on the right. Assume that $r_d = \infty$, $r_o = 0$, and $\alpha \neq \infty$.



- i) Find i_o such that it is in the standard form $i_o = Ai_i - V_L/R_o$ using **exact analysis**. Clearly identify the values for A and R_o . (15)
- ii) Calculate a more accurate R_o via Blackman's Formula for if $r_o \neq 0$. Verify your result by then assuming $r_o = 0$ and comparing to the answer from i). Draw all relevant figures. (10)
- iii) What is this circuit's output voltage compliance? Assume op-amp limits are $\pm V_{sat}$. (10)
- iv) If $i_o = 10$ mA, $V_L = 5$ V, and $R_2 = 200\Omega$, where the current through resistor R_2 is $i_2 = 9.5$ mA (flowing away from ground) and the op-amp is powered via ± 12 V with $i_q = 1$ mA, how much power is being consumed by the op-amp? (10)
- v) From iv): What must the approximate value for R_1 be? (5)

This exam is for:

ECSE-6050

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For each problem: mark the problem number and all subquestions clearly and box your final answer(s). Answers that are not supported by work will not receive credit. Questions with symbolic answers are not expected to be significantly simplified.

All answers will be written on blank paper, provided as necessary.